

Bone augmentation – consultation and treatment

Which approach suits which defect?

An implant needs sufficient supporting bone in the right position. Tooth loss, inflammation or prolonged absence of teeth can leave the ridge too narrow or too low. Bone augmentation aims to improve the conditions for a suitable implant position. This guide explains the main techniques, materials and treatment stages. Only the sections relevant to your individual plan apply to you.

Starting situation	Typical approach in our practice	Implant timing
Small residual defects	Sticky Bone: graft material bound by the patient's own fibrin	Often simultaneous, if stable anchorage in existing bone is possible
Moderate local defects	Semilunar bone shell or titanium-reinforced PTFE membrane	Simultaneous or staged, depending on shape and stability
Larger vertical defects	Shell technique: a stable compartment filled with bone chips	Usually grafting first, implant placement later

This is an orientation to our practice approach, not a rigid size-based rule. Defect shape, blood supply, soft tissues, planned tooth position and implant stability also matter. Techniques may be combined.

Sinus lifting is a special form of augmentation in the posterior upper jaw and is explained separately.

Sinus lift information →

Planning, materials and alternatives

Planning starts with the future teeth

We coordinate the desired restoration with your general dental practice. Examination, scans or models and suitable radiographs show whether sufficient bone is present. CBCT can support three-dimensional planning when the additional information is medically justified; it is not automatically required for every patient.

CBCT: benefits, limitations and costs →

Which materials can be used?

- Your own bone can be collected as local chips or at an additional donor site, often in the posterior lower jaw when a shell is needed.
- Processed donor bone, animal-derived or synthetic substitutes may be suitable alternatives or additions. We discuss the intended material and its origin beforehand.
- Membranes protect the graft compartment from rapidly growing soft tissue. Some dissolve; titanium-reinforced PTFE membranes require planned removal.
- Blood-derived components such as fibrin can bind chips or granules. They do not replace sufficient bone or necessary mechanical stabilisation.

Materials provide a framework or basis for bone formation. Remodelling differs between materials; some substitute particles can remain long term.

Tell us beforehand

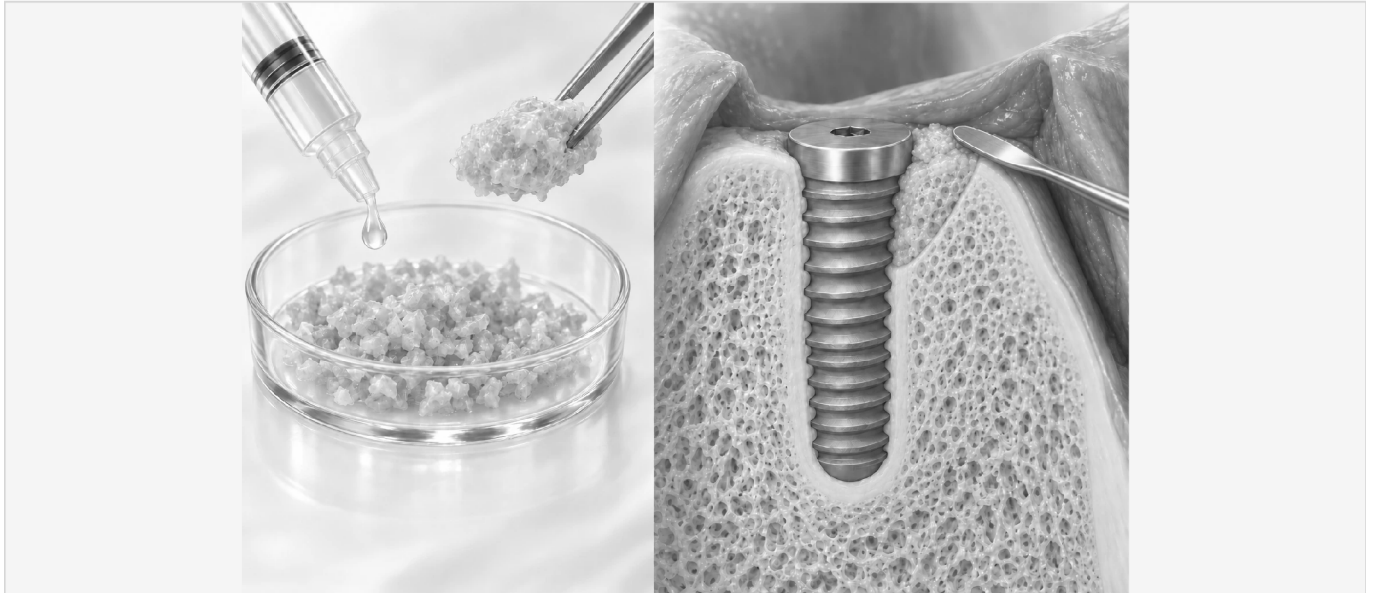
Please report medications, allergies, smoking, diabetes, previous healing problems, bisphosphonate or denosumab treatment and head or neck radiotherapy. Inflammation and poor hygiene should be addressed first. Do not stop regular medication, particularly anticoagulants, on your own.

Alternatives

Depending on the findings, alternatives may include a different implant plan, suitable shorter implants, a bridge, removable teeth or no replacement. We compare benefits, limitations, treatment burden and costs. Grafting cannot guarantee implant success.

Small residual defects: Sticky Bone

Sticky Bone is a cohesive, mouldable mixture of bone chips or graft material and components from your own blood. After a blood sample is taken, a fibrin-containing component is prepared and combined with the graft. Preparation depends on the system used.



A cohesive graft and a limited shoulder defect involving about two to three implant threads in this illustration. AI-generated schematic; symbolic colours, not a patient image.

When do we use it?

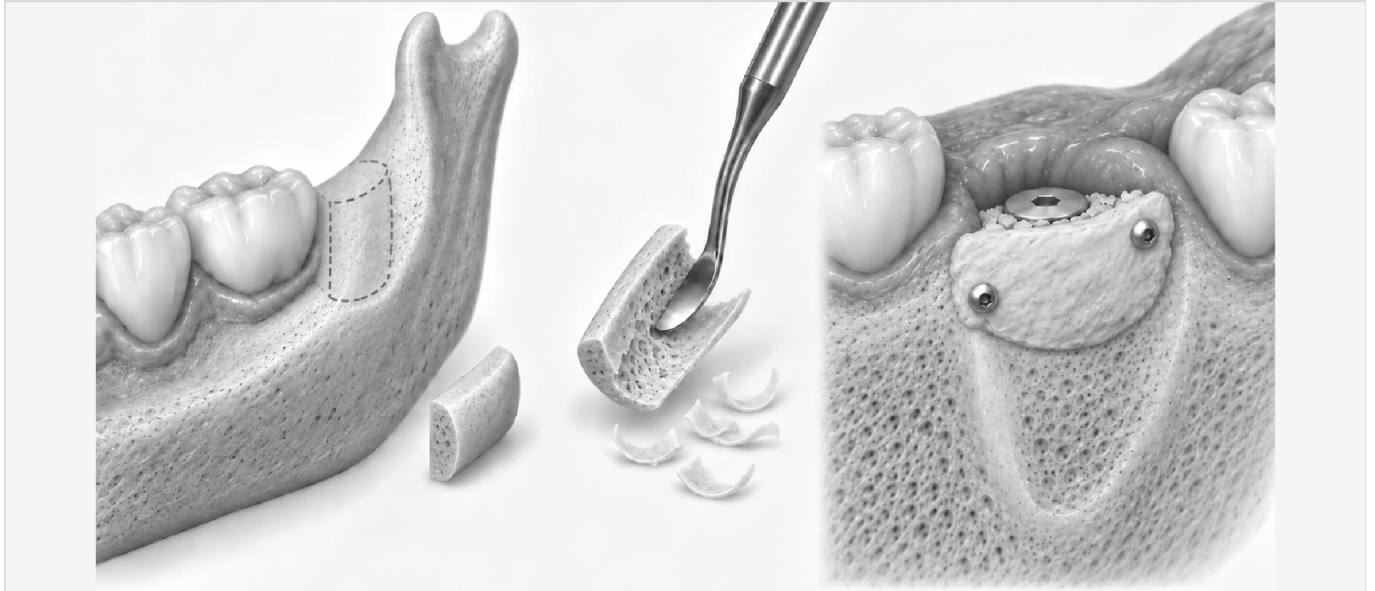
In our practice, mainly to supplement small, local residual defects at the implant shoulder. An implant placed at the same operation must already have sufficient stability in existing bone. The graft cannot securely anchor an otherwise unstable implant.

Benefits and limitations

The fibrin network binds the components and helps adapt the material to the defect. A membrane or additional stabilisation may still be necessary. This mixture alone cannot reliably maintain a large vertical augmentation space. Superior or faster healing compared with every other technique has not been established.

Moderate defects: the semilunar technique

A curved shell made from your own bone covers a local defect. Its shape can help enclose the missing bone area, for example beside an exposed implant shoulder.



Subtly outlined donor region, thinning the bone segment into a shell and securing it over a chip-filled defect. AI-generated schematic; symbolic colours, not a patient image.

How the shell is prepared

- A hollow trephine is used to collect a curved, initially solid bone segment from the posterior lower jaw.
- A bone scraper thins the inside to form a curved shell. The resulting chips can be used for grafting.
- The shell is positioned over the actual defect and secured with small screws. Bone chips fill the space between the defect and shell.

Important considerations

Harvesting creates a second wound. Pain, swelling and altered sensation may therefore also affect the donor site. Simultaneous implant placement is only suitable if stability and defect shape permit; otherwise grafting comes first. Later removal of accessible fixation screws is included in the plan.

Moderate defects: titanium-reinforced PTFE

PTFE is a non-resorbable barrier. An integrated titanium framework helps maintain the intended shape and space over the graft, allowing protected healing. This principle is called guided bone regeneration.



Cross-section: intact bone reaches the implant shoulder on one side; the other side is grafted beneath a reinforced membrane. AI-generated schematic; symbolic colours, not a patient image.

What happens during grafting?

The defect is filled with the planned material, such as bone chips, sometimes combined with a substitute. The membrane is shaped and secured with small pins or screws to prevent movement. The soft tissues are then closed over it with as little tension as possible.

Why is another procedure needed?

PTFE does not dissolve. The membrane and accessible fixation elements are removed after sufficient healing in a planned procedure, sometimes together with implant placement or uncovering. Timing is individual.

If the wound opens or the membrane becomes visible, contact us promptly and do not pull on it. Exposure does not always mean total graft loss. Infection and wound findings determine whether additional treatment or early removal is necessary.

Larger vertical defects: the shell technique

A substantial height deficiency requires a stable compartment above the remaining ridge. Thin bone shells are fixed with small screws and the space is filled with bone chips. Width can sometimes be improved at the same time.



Vertical ridge augmentation in cross-section: fixed shells retain the blue bone chips. The implant is placed later. AI-generated schematic; symbolic colours, not a patient image.

Usually two separate surgical stages

First, bone augmentation: the shells stabilise the graft and the gum is closed over it. **Then, healing:** reassessment is often planned after about four to six months; large grafts or difficult healing may require longer. **Finally, implant placement:** implants are inserted only once adequate supporting bone is present. Their own healing period is additional.

Which shell is used?

The classic technique uses thin shells of your own bone, often harvested from the posterior lower jaw. A processed donor-bone shell may be an alternative in selected cases. We discuss material choice, donor-site needs and differences personally.

For larger vertical defects, simultaneous implant placement is not our usual approach. Staging allows the required bone foundation to develop first. An additional graft may still be needed later.

Treatment and healing

1 · Prepare and decide

We agree the technique, material, anaesthesia, implant timing and temporary restoration. You receive an individual cost plan. Removable teeth must be adjusted to keep pressure off the surgical site. Allow time for reviews and the final restoration.

2 · Operate and protect

After effective anaesthesia, the defect is exposed and grafted. Shells or membranes are secured where necessary, and wound closure protects the graft. Unexpected findings can require a changed or staged plan.

3 · Review and heal

Swelling, bruising and restricted mouth opening may occur. Reviews are arranged individually. Sutures are often removed after one to two weeks, sometimes later with special wound closures. Avoid chewing pressure or pressure from temporary teeth. The gum can look healed while bone maturation still takes months.

4 · Implant placement or uncovering

After a separate graft, we assess healing clinically and with appropriate imaging when needed. Membranes and fixation elements are removed as required. We then decide whether to place implants. Implants placed during grafting may later require uncovering after successful healing.

5 · Restoration and long-term care

Your general dental practice restores the healed implants. Daily cleaning and regular reviews remain essential. Grafting does not permanently prevent inflammation or future bone loss.

[Dental implants: consultation, treatment and uncovering →](#)

How long does everything take?

Staged augmentation can add several months to implant healing and restoration time. All times are estimates and depend on the individual findings, not guaranteed completion dates.

Anaesthesia and comfort options

Anaesthesia and comfort options

Local anaesthesia is standard. You remain awake and may feel pressure or movement. If you feel pain, we stop and give additional anaesthetic. Sedation and general anaesthesia require a separate assessment and consent discussion.

Method	Procedure and important rules	Costs
Local anaesthesia Standard	The area is numbed; you remain awake. Public transport home is possible; individual fitness to travel is assessed.	Included in the surgical treatment plan.
Oral sedation Lorazepam (Tavor)	A tablet usually about 60 minutes beforehand reduces anxiety without guaranteeing sleep. Vital signs are monitored. An escort must collect you from the practice. No driving for 24 hours.	About €80 privately paid.
Intravenous sedation	Not routinely offered for implant placement in our practice: small components and coughing or gagging may create an aspiration risk. Only after individual assessment with the anaesthesia team. If agreed: Tuesdays, monitored recovery, collection from the practice by an escort mandatory; no driving for 24 hours.	Exceptional cases only; agree costs beforehand.
General anaesthesia	Deep sleep with a secured airway, usually a breathing tube passed through the nose. Anaesthesia team available on Tuesdays, with monitored recovery. An escort must collect you from the practice; no driving for 24 hours.	Time-dependent; usually about €500–800 additional anaesthesia costs.

Local anaesthesia is sufficient for most straightforward implant procedures. Follow the anaesthesia team's specific fasting instructions for IV sedation or general anaesthesia. Our usual schedule is no food after 10 pm and no drinks after 6 am; different individual instructions take priority. Take medication only as agreed.

Anaesthetic risks

Local anaesthesia may cause circulatory reactions, bruising, allergic reactions and rarely nerve irritation. Sedation and general anaesthesia can affect breathing and circulation and cause nausea, vomiting or aspiration; serious complications are rare but possible. Monitoring, appropriate selection, fasting and escort arrangements are therefore important.

Risks, limitations and costs

General and technique-specific risks

- Pain, swelling, bruising, bleeding, infection, restricted mouth opening and delayed wound healing.
- Wound opening, exposed membranes or shells, loose fixation elements, and partial or complete graft loss. Early removal or further surgery may be needed.
- Loss of graft volume or insufficient bone gain, requiring further grafting, a changed implant plan or a different restoration.
- Damage to nearby teeth, vessels or nerves. Numbness, tingling or altered sensation in the lip, chin or tongue may be temporary and rarely permanent.
- Additional pain, infection or altered sensation at a donor site; very rarely weakening or fracture of the jaw.
- Depending on location in the upper jaw, sinus opening or inflammation. Simultaneous implants also carry implant-specific risks, including failure to integrate or implant loss.

Smoking, untreated periodontitis, inadequate hygiene, poorly controlled diabetes and certain medicines increase risk. Bone-active medication or prior radiotherapy can make impaired healing and jaw osteonecrosis particularly relevant. Individual risks are discussed personally.

Costs

Costs depend on extent, technique, materials, harvesting, anaesthesia and the number of procedures. An individual estimate should also cover planned later procedures. Complications requiring additional treatment may lead to further costs.

In Germany, grafting for implants is usually a privately agreed service for patients with statutory insurance; exceptions and reimbursement require individual assessment. Private cover depends on your policy. If CBCT is needed, our charge is €180–280 depending on the diagnostic question.

Your decision

You can ask questions, take time to decide and consider alternatives. This guide supplements personal informed consent and does not imply consent to every technique described.

Supporting healing

- **Keep pressure off the graft:** do not bite onto the freshly operated ridge. Choose soft food and avoid the site. Wear removable teeth only as approved and have pressure points checked promptly.
- Maintain careful hygiene elsewhere; clean the wound only as instructed. Use prescribed antiseptic rinses for the agreed period without vigorous rinsing.
- Avoid smoking and other nicotine products. Resume exercise according to the procedure and clinical advice.
- Cool intermittently without direct ice contact. Do not pull on the lip, wound, stitches or visible materials.
- Take medicines as prescribed. Avoid extra aspirin as a painkiller, but do not independently stop prescribed aspirin or other blood-thinning medication.
- Attend reviews. Seek advice for increasing pain or swelling after initial improvement, fever, pus, persistent bleeding, new altered sensation or a visible membrane.

Detailed recovery advice after implant surgery →

Practice: 06151 786540. After surgery outside consultation hours: +49 151 10437450. If unavailable: dental emergency service 01805 60 70 11. For a life-threatening emergency, breathing difficulty, severe difficulty swallowing or rapidly increasing swelling of the floor of the mouth, neck or tongue: 112.

Common questions

Do I choose the technique myself? We recommend an approach based on your findings and discuss the reasons and alternatives.

Does all material become my own bone? Remodelling depends on the material; some substitute particles can remain long term.

Is a larger graft automatically better? No. We plan the amount required for a suitable restoration.



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